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Project Synopsis Template (2025-26) - Sem VII

NiyatKalpa- A timely solution for Medicines

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Abstract

India faces two intersecting challenges in the healthcare sector — the widespread wastage of unused or expired medicines and the persistent lack of affordable access to essential medications for low-income populations. Pharmaceutical waste is increasing annually, and the lack of proper disposal practices poses significant risks to both environmental and public health. At the same time, a considerable portion of the population struggles to afford or access basic drugs due to price, distribution, or geographic limitations [1], [2].

This growing imbalance between surplus and shortage calls for innovative, scalable, and sustainable interventions. Digital healthcare systems, AI-based assistance, and mobile platforms have demonstrated their potential in improving healthcare accessibility and awareness [6], [7]. To address these dual concerns, the proposed project explores the development of a mobile-based digital platform focused on redistributing near-expiry medicines while providing intelligent pharmaceutical guidance, thereby improving health equity and reducing waste.

Introduction

Access to essential medicines is a fundamental human right, yet many individuals across India still lack consistent access due to affordability, distribution, and infrastructure gaps. According to the World Health Organization, up to 30% of people in developing countries face limited or irregular access to basic drugs [10]. In India, out-of-pocket expenditure remains the leading cause of medical poverty, contributing to health disparities and delayed treatments for millions. While national initiatives like the *Ayushman Bharat Digital Mission* have made strides in digitizing health records and expanding insurance coverage, they fall short in solving the problem of medicine affordability at the grassroots level [12].

On the other hand, unused and expired medicines continue to accumulate in households, hospitals, and retail pharmacies. Studies indicate that the majority of people do not follow safe disposal practices, with over 70% of expired drugs ending up in trash bins or flushed into water systems [2], [3]. This not only causes environmental pollution but also raises the risk of antimicrobial resistance and unregulated drug usage. There is currently no standardized national framework in India for collecting, sorting, or safely redistributing these unused medications. This disconnect between

abundance and lack forms the foundation of the current research and development efforts toward a corrective solution.

Problem Statement

Despite advancements in digital healthcare and pharmacy infrastructure, India continues to suffer from systemic inefficiencies in the distribution and disposal of medicines. The absence of an efficient reverse logistics model for collecting and repurposing unused or near-expiry medications leads to enormous wastage each year. A 2020 survey across Delhi NCR found that 87% of respondents stored unused medicines, and nearly three-quarters discarded expired drugs without medical consultation or eco-friendly methods [2]. These trends reflect not only a public health oversight but also a missed opportunity to repurpose valuable pharmaceutical resources.

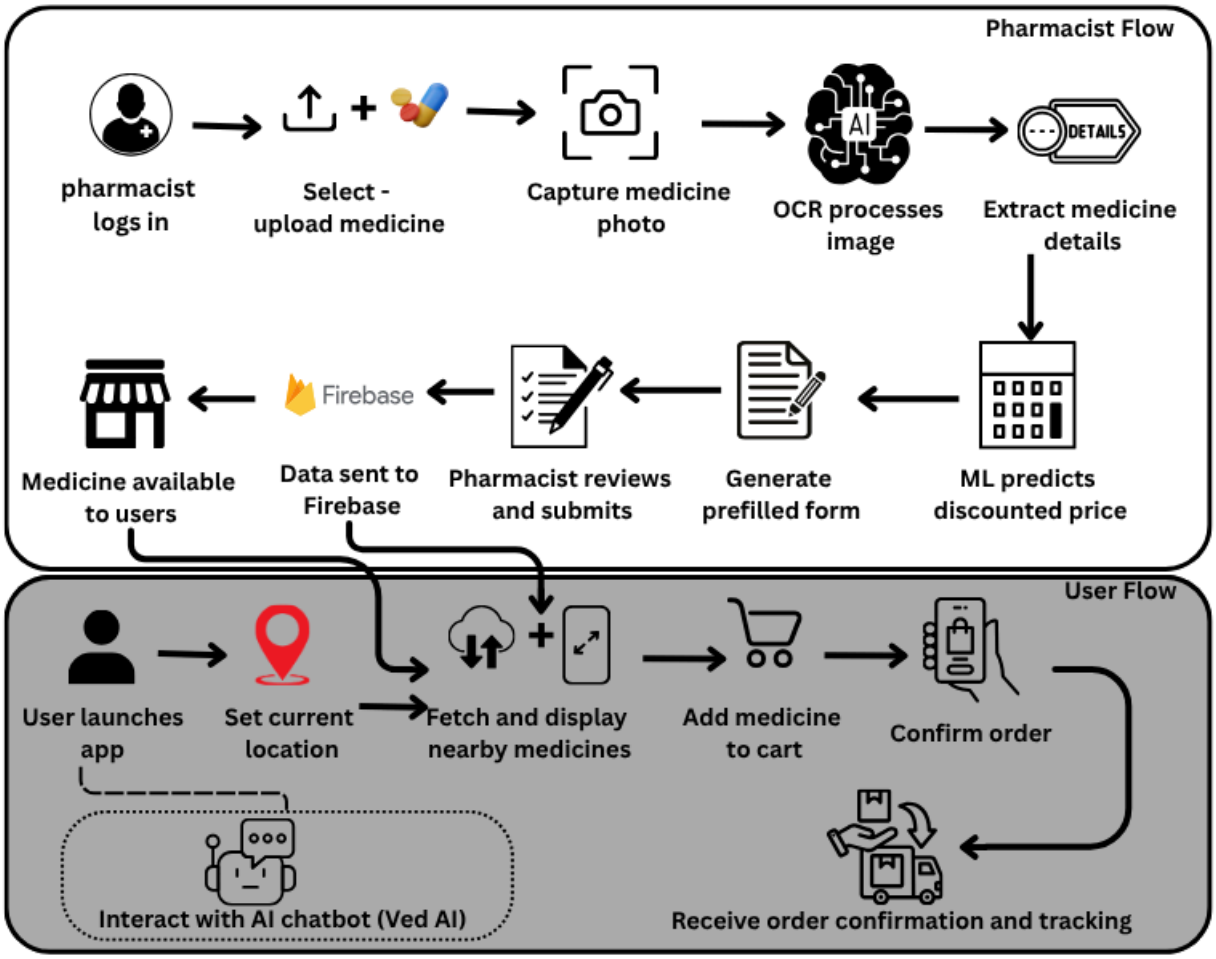
Simultaneously, a significant segment of India's population — particularly in rural and economically weaker areas — cannot afford to purchase the very medications being discarded elsewhere. The lack of affordable options exacerbates medical neglect, increases the disease burden, and prolongs recovery periods. While some non-governmental and informal networks attempt to bridge this gap, there exists no robust, tech-enabled national mechanism that connects surplus medicines with people in need [4], [5]. The convergence of these two systemic problems demands a digitally enabled, sustainable, and ethically managed solution.

Proposed Solution

The proposed solution is a mobile-based application called *NiyatKalpa* that distributes near-expiry, yet safe-to-use, medicines to individuals from low-income or underserved communities at highly subsidized rates. The core idea is to connect local medical stores — which typically dispose of such surplus drugs — with end-users who are willing to buy them at reduced prices. This creates a dual benefit: preventing pharmaceutical waste and promoting accessible healthcare. Users can search for discounted medicines based on their location, check expiry dates, and place orders through a secure, easy-to-use interface. Additionally, the app incorporates a novel feature allowing stakeholders (pharmacy staff or users) to capture a photo of the medicine packaging, where an image processing and OCR-based mechanism scans and extracts the expiry date automatically. This automation reduces manual entry errors and improves the reliability of expiry tracking.

NiyatKalpa will also feature an AI-powered chatbot named Ved AI, which provides real-time information about medicines including dosage guidelines, side effects, contraindications, and usage precautions. This ensures users are empowered with medical knowledge before purchase. The app will integrate **Firebase** for user authentication, database management, and cloud storage, while **React Native** will ensure smooth cross-platform deployment. Key modules include medicine listing, smart search, order management, AI interaction, pharmacy locator via **Google Maps API**, and the intelligent expiry date scanning tool. Future upgrades will introduce features like digital prescription uploads, secure payment gateway integration, multilingual support, and doctor consultations. The solution prioritizes **sustainability, affordability, and intelligent access**, leveraging modern mobile and AI technologies to solve real-world health challenges.

Methodology / Block Diagram :



The development methodology for NiyatKalpa follows a modular, layered architecture built on a mobile-first technology stack using React Native for the frontend and Firebase for backend services. The frontend is implemented using React Native with TypeScript,

which ensures type safety, a better developer experience, and cross-platform support for both Android and iOS devices. The backend stack leverages Firebase Authentication for secure user and pharmacist login, Cloud Firestore for real-time data storage, and Cloud Functions for backend logic and automated tasks.

The system is divided into two primary user flows: the pharmacist workflow and the end-user (medicine seeker) workflow. Pharmacists access the app's medicine upload interface, where they only capture a photo of the medicine packaging using their device camera. This image is processed using an integrated Optical Character Recognition (OCR) system — such as Google ML Kit or Tesseract — to automatically extract critical fields like medicine name, batch number, manufacturer, and expiry date. Importantly, the expiry date is locked (non-editable) after extraction to preserve data accuracy.

Following this, the application uses an AI-based pricing engine, which evaluates the remaining shelf life (days until expiry) and applies a smart discounting logic to recommend a dynamic selling price. A pre-filled submission form is shown to the pharmacist, allowing them to review the extracted details and confirm the listing. Once submitted, this data is stored in Firebase Firestore, and the listing becomes visible to nearby users, filtered via geolocation tagging.

On the user side, individuals log into the app and set their location using the integrated Google Maps API. They are then shown a curated list of near-expiry medicines available in nearby pharmacies, along with real-time data like expiry date, available quantity, and discounted price. For additional guidance, users can interact with Ved AI, a built-in AI chatbot powered by Zapier automation and NLP logic, which answers medicine-related questions (e.g., dosage, contraindications, usage tips) in simple, natural language.

This architecture offers a smart, sustainable, and user-centric approach to medicine redistribution — one that minimizes human error, reduces medical waste, and supports affordable healthcare delivery through intelligent automation.

Hardware , Software and tools Requirements

Hardware Requirements:

- **Smartphones** (Android/iOS) with camera capability: For users and pharmacy staff to run the NiyatKalpa app and capture medicine package images for expiry scanning.
- **Development Machines**: Laptops or desktops with minimum 8 GB RAM and SSD storage for coding, testing, and deployment.

- **Server Infrastructure**: Cloud-based hosting via Firebase services, no dedicated physical server required.

Software Requirements:

- **React Native**: For cross-platform mobile app development using JavaScript/TypeScript.
- **Firebase**: Including Firebase Authentication (for secure user login), Cloud Firestore (for real-time database), Cloud Functions (for backend logic), and Firebase Storage.
- **Google Maps API**: For location-based pharmacy discovery and navigation.
- **OCR Libraries/SDKs**: Such as Google ML Kit OCR or Tesseract OCR integrated into the app for expiry date recognition from images.
- **Zapier or Equivalent**: To automate AI chatbot workflows for Ved AI's pharmaceutical guidance.
- **Version Control**: Git and GitHub/GitLab for source code management.
- **Development Tools**: Android Studio, Visual Studio Code, Firebase Emulator Suite for local testing and debugging.

Proposed Evaluation Measures

Evaluation Metric	Description	Method of Measurement
User Onboarding Experience	Ease of app installation, signup, and navigation for first-time users.	User surveys and task completion time in usability testing.
Medicine Data Accuracy	Correctness of expiry date extraction via OCR and reliability of medicine details.	Accuracy percentage in OCR tests against manually validated data.
AI Chatbot Response Quality	Relevance and correctness of Ved AI's medical advice and information.	Expert review and user satisfaction surveys.
Pharmacy Locator Accuracy	Accuracy of displayed nearby pharmacies and real-time stock availability.	Comparison of app data with actual pharmacy inventory and location data.

Order Fulfillment Rate	Percentage of orders successfully completed without errors or delays.	App analytics tracking order statuses over a test period.
Medicine Waste Reduction Potential	Estimated volume of medicines redistributed vs. wasted over a pilot timeframe.	Periodic reports and pharmacy collaboration data.
System Performance and Scalability	App responsiveness, load handling with concurrent users, and error rates.	Performance testing using Firebase analytics and stress tests.

Conclusion

NiyatKalpa addresses the pressing public health and environmental challenges arising from the simultaneous prevalence of unused medicines and limited access to affordable drugs in India. By leveraging mobile technology, cloud computing, AI-driven assistance, and innovative image-based expiry scanning, the platform aims to build a sustainable ecosystem that empowers users and pharmacies alike. This dual approach not only reduces pharmaceutical waste but also enhances equitable healthcare accessibility for marginalized populations.

The project's modular, scalable architecture ensures adaptability to varied user demographics and geographic contexts. Future enhancements such as integrated payment gateways, prescription management, multilingual support, and expanded AI capabilities will strengthen the platform's impact. NiyatKalpa embodies a socially responsible, technology-driven solution aligned with national and global health priorities, marking a significant step toward smarter, more sustainable healthcare delivery in India.

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